

MedKayfab Laboratory Medicine Wrestling: The Numerical Noir & The Biomarker Betrayal.

Uncle FrizzleBob's Krazy Kayfabizzaro Komik
Kards for Narrative Medical Mnemonic Fun



MedKayfab!

What the
FLUFF?!

Diagnosis?!

CRITICAL
HIT!

The numbers never
lie, but they often
speak in riddles.
Stay fluffy!



CRITICAL
HIT!

BIOMARKER
BLAST!

NUMERICAL
SMACKDOWN!



Sodium-the-Volume-Lord (Na^+)

135-145 mmol/L
Extracellular cation.



POWER: The Water-Warp.
Pulls Water-Tokens
wherever he goes.

Too high (Hyper-Nate) = brain cells shrink.
Too low (Hypo-Nate) = cells swell and explode.

LORE: The Osmotic-Anchor. Fixing Hypo-Nate too fast summons
Osmotic-Demyelin-Dan (paralysis).

Calcium-the-Concrete (Ca^{++})

8.5-10.5 mg/dL
Ionized vs. Total.



POWER: The Membrane-Stabilizer. Keeps Nerve-Gates steady.
Hypo-Cal = Gates flap (Tetany/Seizures).
Hyper-Cal = Gates freeze (Stones/Noans).

LORE: Regulated by Parathyroid-Pete. If Mag-the-Magnet is low,
Calcium won't rise.

Potassium-the-Spark-Plug (K^+)

3.5-5.0 mmol/L
Intracellular cation.



POWER: The Voltage-Trigger. Controls the Resting-Reset of
heart wires. OFF-balance? The heart quivers.

LORE: Hyper-K-the-Killer creates Peaked-T-Wave-Mountains.
Hypo-K-Rarl digs U-Wave-Valleys. Razor-thin margin of error.

Mag-the-Magnet (Mg^{++})

1.5-2.5 mg/dL
Co-factor for 300+ enzymes.



POWER: The Gatekeeper's-Helper. The Universal-Relaxer
holding the Potassium/Calcium gates shut.

LORE: Low Mag triggers Tersades-the-Twist.
The Ignored-Electrolyte causing Mystery-Glitches.

KayfaBINGO!

Tersades

Hemoglobin-the-Carrier (Hb)

12-16 g/dL
Oxygen capacity.



POWER: The Oxygen-Tank. Determines available Breath-Fuel.
Low Hb (Anemia-Annie) = fainting from Energy-Paucity.

LORE: The Cargo-Ship. Without Iron-Bolts, the ship can't carry oxygen-crates.

Platelet-the-Plug (Plt)

150-450k
Primary hemostasis.



POWER: The Hole-Sealer. First-Responders to a Leak-Crisis.
Low count (Thrombo-the-Thin) = bleeding from scratches.

LORE: Too high = Clot-the-Blocker.
Beware HIT-the-Hunter in the ICU!

WBC-the-War-Count

4-11k
Neutros, Lymphs, Monos, Eos, Basos.



POWER: The Riot-Scanner. High (Leuko-the-High) = Infection-Incursion. A Left-Shift means the Bone-Marrow-Factory deployed Untrained-Rookie-Soldiers.

LORE: Neutrophils fight bacteria. Eosinophils fight Parasite-Punks and Allergy-Allys.

MCV-the-Size-Gage

80-100 fL
Mean Corpuscular Volume.



POWER: The Cell-Architect. Tells the Detective if Red-Cells are tiny (Micro-Mitch) or giant (Macro-Max).

LORE: Micro-Mitch = Iron-Deficiency.
Macro-Max = B12-or-Folate-Famine.

BLÖDSINN!



Creatinine-the-Waste-Marker

0.6-1.2 mg/dL
GFR estimate.

09



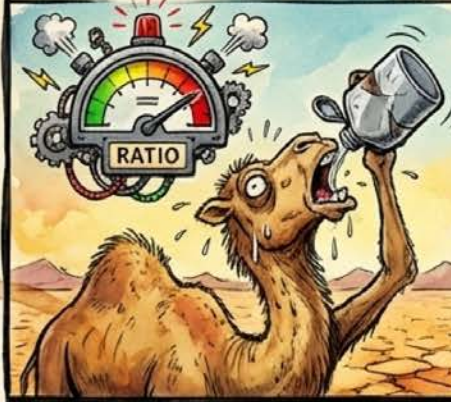
POWER: The Filter-Score. The Trash-Count in the blood. Rising score = Kidney-Filters are clogged/failing.

LORE: A lagging indicator. If Krea doubles, 50% of the Kidney-Gears have already seized up.

BUN-the-Urea-Messenger

7-20 mg/dL
Blood Urea Nitrogen.

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POWER: The Dehydration-Signal. BUN-to-Krea-Ratio > 20 means the wrestler is Fluid-Starved.

LORE: Rises when thirsty, or if Hank-the-Hemorrhage is hiding in the GI tract (Protein-loading).

GFR-the-Flow-Rate

> 90 mL/min
The ultimate kidney stat.

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POWER: The Clearance-Clock. How fast the body can Flush-the-Toxins and Clear-the-Meds.

LORE: Calculated from Krea/Age/Gender. If GFR < 15, it's Dialysis-Time.

Cystatin-C-the-Pure-Filter

Alt to Creatinine
Unaffected by muscle.

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POWER: The Precise-Scan. For wrestlers too Sarcopenic-Thin or Bodybuilder-Buff for a standard Krea-check.

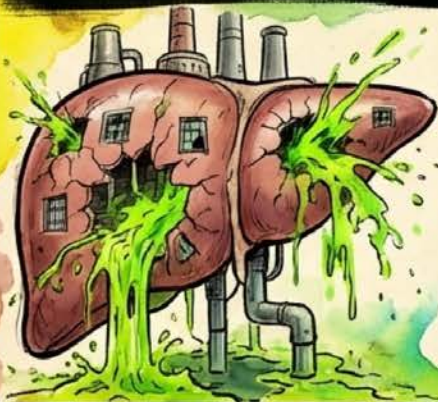
LORE: A stable Trash-Marker produced constantly by all nucleated cells.



Call Doc Frizzle!

ALT-the-Liver-Leak

< 40 U/L
Specific to liver cells.



POWER:
The Cell-Burst-Signal.
High ALT =
Liver-Cells are
smashed, leaking
their Chemical-Guts
into the blood.

LORE:
The Liver-Sentry.
Thousands of units =
Acute-Toxic-Assault
or Viral-Vandalism.

AST-the-Muscle-Misfit

< 40 U/L
Liver, Heart, Muscle.



POWER:
The Multi-Damage-
Marker. Less
specific than ALT.
If AST is much
higher than ALT
(2x), think Al-the-
Alcoholic.

LORE: Also spikes in
Heart-Halt and
Muscle-Meltdown
(Rhabdo).

Albumin-the-Pressure-Magnet

3.5-5.0 g/dL
Half-life 20 days.



POWER:
The Chronic-Status-Bar.
Built by the liver; a
low score means the
Liver-Factory has been
failing for weeks.

LORE:
Low Albumin causes
Edema-the-Swelling
because the blood's
Water-Magnet is gone.

Bili-the-Bile-Builder

Direct vs
Indirect Bilirubin.



POWER:
The Jaundice-Glow.
High Indirect =
Red-Cells exploding.
High Direct = Bile-
Ducts are blocked.

LORE:
The Waste-Product of
hemoglobin. High Direct
Bili causes Dark-
Urine-and-Pale-Stool.

17 Troponin-the-Heart-Scream

hs-Tn
Peaks at 12-24h.



POWER: The Infarct-Verdict.
The Gold-Standard proof that
Cardio-Carl's heart cells are
actively dying.

LORE: The Molecular-Skeleton.
If it's in the blood, the cell
membrane has been fatally breached.

19. CK-MB-the-Re-Infarct-Finder

Creatine Kinase-MB
Normalizes 48-72h.



POWER: The Timer.
Disappears fast. Used to
see if the wrestler is
having a second heart
attack right after the
first.

LORE: Troponin stays
high for days; CK-MB is
the Short-Term-Witness.

18 BNP-the-Stretch-Signal

Brain Natriuretic Peptide
< 100 pg/mL.

POWER: The Flood-Warning.
When heart-walls Stretch
from too much fluid, BNP
yells at kidneys to
Dump-the-Water.

LORE: Differentiates
Lung-Moni-Pneumonia
from Heart-Failure-
Flood.



20. D-Dimer-the-Clot-Debris

Fibrin degradation
High sens, low spec.

POWER: The Clot-Rule-Out.
Low D-Dimer = NO clot. High
= Something is breaking
down (Infection, Trauma,
or Clot).

LORE: The Scraps of a
fibrin clot. It is a Screen,
never a Diagnosis.



Diagnosis??



21

PT/INR-the-Warfarin-Tracker

Prothrombin Time
Extrinsic/ Factor VII.

POWER: The Thin-Check. Monitors Warf-the-Old-Guard. High INR = High Bleed-Risk.
LORE: Liver failure raises INR because the sick liver can't build Factor-7-the-Sentry.

22

aPTT-the-Heparin-Hitch

Activated Partial
Thromboplastin Time.

POWER: The Speed-Limit. Monitors Heparin-the-Heavy.
 Too high = blood acts like Water-Tokens.
LORE: Measures the Intrinsic-and-Common-Highways.

23

Fibrinogen-the-Glue-Supply

200-400 mg/dL
Acute phase reactant.

POWER: The Resource-Count. Without it, Platelet-Plugs can't stick. In DIC-the-Disaster, the supply is Consumed to zero.
LORE: The vital Raw-Material for the Fibrin-Mesh.

24

Anti-Xa-the-Modern-Monitor

Monitors LMWH or
Rivaroxaban.

POWER: The X-Ray-Vision. Tracks Factor-X-Inhibitors when standard PT/aPTT are completely blind to them.
LORE: Used in Obese-Wrestlers or Renal-Fail-Rays to verify dose safety.



TSH-the-Master-Signal

Thyroid Stimulating Hormone
0.5-5.0 mIU/L.



POWER: The Feedback-Loop. High TSH = Brain screaming at a Sleepy-Theo-Thyroid. Low TSH = Thyroid is Over-Revving.
LORE: Most sensitive thyroid test. The Pituitary-Sentry is always watching.

HbA1c-the-Three-Month-Memory

Glycated Hemoglobin
Goal < 7%



POWER: The Cheat-Detector. Reveals Average-Sugar-Stats over 90 days. You can't hide a Pizza-Binge from A1c!
LORE: Sugar permanently glues itself to the Red-Cell-Inventory for its full 120-day lifespan.

Cortisol-the-Stress-Stat

AM (High) vs PM (Low)
Synacthen test.



POWER: The Survival-Fuel. Zero = Addison-the-Crash (Shock). Too high = Cushing-the-Curb (Obesity/Thin skin).
LORE: Fight-or-Flight. Caveat: High ICU stress Artificially spikes this stat.

C-Peptide-the-Insulin-Twin

Secreted 1:1 with
endogenous insulin.



POWER: The Origin-Finder. Proves if blood insulin was Home-Made (Pancreas) or Injected (Syringe).
LORE: Type 1 Diabetics have ZERO C-Peptide. Insulinoma-Ian has massive amounts.

KayfaBIZARRO!

29 ABG-the-Blood-Gas-Bella

pH, pCO₂, pO₂, HCO₃⁻
Arterial.



POWER: The Living-Truth. Tells exactly how well Lung-Balloons exchange air and if blood is Sour-with-Acid.
LORE: ROME (Respiratory Opposite / Metabolic Equal). Low pH + High CO₂ = Lungs-are-Lazy.

31 Anion-Gap-the-Missing-Acid

Na - (Cl + HCO₃)
Normal 10-14.



POWER: The Mystery-Acid-Finder. High Gap = a Hidden-Acid-Saboteur is in the blood (Ketoacids, Lactic, Toxins).
LORE: MUDPILES. The Gap tells you precisely what kind of acid is ruining the match.

30 Lactate-the-Oxygen-Debt

> 2 mmol/L
Sepsis marker.



POWER: The Starvation-Counter. High Lactate = Cell-Engines are starving for oxygen, burning Emergency-Fuel.
LORE: The toxic Smoke billowing from a Tissue-Fire.

32 Carboxy-the-Carbon-Copy (CO-Hb)

Carbon Monoxide
Pulse-ox is blind!



POWER: The Silent-Suffocator. CO hitchhikes a ride on the Red-Cell-Inventory, permanently blocking oxygen from boarding.
LORE: The Cherry-Red wrestler. The standard SpO₂-Sensor is totally fooled into thinking blood is 100% oxygenated.

Stay fluffy!

GW

Georg v. Westphalen

33

CRP-the-Acute-Alert

< 5 mg/L
Liver produced
IL-6 driven.



POWER: The Fire-Alarm. Rises fast during Sam-Sepsis or a Joint-Riot. Confirms the Battle-is-Active.
LORE: An Acute-Phase-Reactant. Doesn't say where the fire is, just that the Immune-Army is mobilizing.

34

ESR-the-Sed-Rate

Erythrocyte
Sedimentation Rate
mm/hr.



POWER: The Slow-Burn-Tracker. Spikes in chronic Auto-Immune-Wars like Lupus-Lucy. The Old-School-Inflammation-Clock.
LORE: Inflammation makes red cells stick together and Sink-Past in a test tube.



KayfaBONGO!

Procal-the-Bacteria-Beacon

Procalcitonin
Specific for bacteria.



POWER: The Antibiotic-Key. High = Bacterial-Punks are present. Low = it's just a Viral-Vandal.
LORE: The ultimate weapon to help the team stop Antibiotic-Overuse.

Blood-Culture-the-Clan-Finder

Aerobic & Anaerobic
2 sets, diff sites.



POWER: The Positive-Identification. Tells the team the exact Name-and-Weapon of the bacteria in the Blood-Highway.
LORE: Sterile-Technique is vital. If Skin-Steward-Staph falls into the bottle, it's a False-Alarm.

37 PSA-the-Prostate-Picket

Prostate Specific Antigen
/ < 4 ng/mL.



What the
FLUPPY!



POWER: The Gland-Guard. High PSA = Proetate-Towers are inflamed (Prostatitis) or under siege by Malignant-Max.
LORE: Not 100% specific. A massive BPH-Benign-Gland also raises the score.

38 CEA-the-Colon-Monitor

Carcinoembryonic Antigen.



POWER: The Recurrence-Radar. Not for finding cancer initially, but watching if it returns after the Blade-Brigade removes it.
LORE: A rise after surgery = the Shadow-Malignancy is creeping back.

39 CA-125-the-Ovarian-Alert

Elevated in 80% of
ovarian cancers.



POWER: The Peritoneal-Patrol. A Follow-up tool. Also triggers for Endo-Bnid and Fibroid-Flona.
LORE: Any physical Irritation of the peritoneal lining can artificially raise this score.

40 AFP-the-Liver-Tumor-Track

High in HCC &
Germ cell tumors.



What the
FLUPPY!

POWER: The Mass-Classifer. If Liver-the-Limp has a lump + high AFP, it's likely Malignant-Max.
LORE: Dual-uae: Also the marker for Petal-Neural-Tube-Leaks in the Gyn-deck.

41 UA-the-Dipstick

Leuks, Nitrites, Blood,
Protein, Glucose, Ketones.



POWER: The Quick-Urine-Scan. Reveals a Bladder-Riot (UTI), Sugar-Leak (Diabetes), or Filter-Failure (Proteinuria).
LORE: Nitrites + Leukocytes = Bacteria-the-Punks are absolutely present.

42 Micro-Albumin-the-Early-Warning

30-300 mg/day
Urine alb/krea ratio.



POWER: The Diabetic-Kidney-Sentinel. Finds the Tiny-Protein-Leaks long before the standard dipstick can see them.
LORE: The very first clinical sign of Diabetic-Nephropathy.

43 Stool-the-Fecal-Occult (FIT)

Feecal
Immunochemical Test.



POWER: The Hidden-Blood-Finder. Locates Invisible-Hank-the-Hemorrhage in the gut (early sign of Apple-Core-AI).
LORE: Only detects human blood. No Red-Meat-Interference to cause false alarm!

44 CSF-the-Meningitis-Map

Glucose, Protein,
WBC count, Pressure.



POWER: The Brain-Bunker-Report. Differentiates Bacterial-Riot (Low sugar, High protein) from Viral-Vandalism (Normal sugar).
LORE: Drawing the sacred Liquor of the soul to check the inner sanctum.

45 Ammonia-the-Brain-Fog (NH_3)

Protein waste
Cleared by the liver.



POWER: The Toxic-Cloud. If Liver-the-Limp fails, Ammonia builds up and triggers Encephalo-Ed (Confusion/Coma).
LORE: The chemical culprit behind Asterixis (The Flapping Hand-Tremor).

46 Lipase-the-Pancreas-Puncture

More specific than Amylase
Peaks in 24h.



POWER: The Autodigestion-Alarm. When Lipase is $>3\times$ normal, the Pancreas-Factory is literally Bating-Itself.
LORE: The definitive hallmark of Acute Pancreatitis.

47 Uric-Acid-the-Needle-Count

High in Gouty-George
& Tumor-Lysis.



POWER: The Crystal-Slayer. High levels precipitate into Needles that ruthlessly stab the joints and the Kidney-Filters.
LORE: The painful end-product of Purine metabolism.

48 Ferritin-the-Iron-Vault

Best marker for iron
stores / Acute phase.



POWER: The Inventory-Check. Proves if the Iron-Bolts are truly missing (Anemia-Annie) or just hidden away.
LORE: Caveat: During a Systemic-Riot, the body locks the Iron in the vault. Ferritin will look Falsely-High.

49 Digoxin-the-Narrow-Window Therapeutic 0.5-2.0 ng/mL Monitor K+.



POWER: The Cardiac-Gamble. Level > 2.0 = Dig-Tox-Dan appears, causing Yellow-Vision and heart blocks.

LORE: Hypokalemia makes the heart wires Hyper-Sensitive to Digoxin's poison.

51 Lithium-the-Salt-Level Therapeutic 0.6-1.2 mmol/L Renal.



POWER: The Mind-Stabilizer. Too low = Barry-Bipolar swings. Too high = The-Shakes and immediate Renal-Failure.

LORE: Competes directly with Sodium in the Kidney-Filters.

50 Acetaminophen-the-Nomogram Rumack-Matthew 4-hour level.



POWER: The Liver-Destiny. Predicts if the wrestler survives the overdose or if the Liver-Factory is mathematically doomed.

LORE: Treatment relies entirely on the N-Acetyl-the-Cysteine-Rescue.

52 Ethano-the-Al-Level Legal vs Toxic Zero-order kinetics.



POWER: The CNS-Depressor. Decodes why the wrestler is unconscious. Is it The-Drink or The-Stroke?

LORE: Cleared at an absolute, unchangeable constant rate of 15-20 mg/dL per hour.

$$\text{Clearance} = \frac{\text{Lithium} - 1^2}{\log(V_u - V_1 \cdot 6 \cdot 10^3)} \quad k_{el} = \frac{V_u \cdot C_1 - C_2}{t}$$

$$\text{Lithium} = 12.2 \frac{\text{clearance}}{\text{heavy}} \times 0.19$$

$$\text{Lithium} = \frac{\text{clearance}}{12.095 \text{ absolute}} \times \frac{3.05}{15-20 \text{ mg/dL}}$$

$$L_{el} = \frac{10314}{113.23} \quad \text{GW Georg v. Westphalen}$$

Call Doc Priexlo!

$$\text{Digoxin} = \frac{(1.005 \times C_1) - 0.7V_1^2}{(0.03 \times C_1) + k_0}$$

$$Cl = \frac{(V_u \times C_1 - V_1 \times C_2)}{t} \quad k_{el} = \frac{0.693}{t_{1/2}} \quad k_{el} = \frac{K_{el}}{t_{1/2}}$$

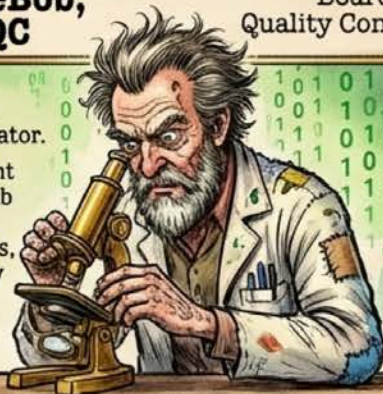
Doc FrizzleBob, Master of QC

Board Certified /
Quality Control Master.

POWER:

The Data-Validator.

LORE: The patient
is the text; the lab
is the subtext.
Trust the numbers,
but always verify
the clinical fit!
Stay fluffy!



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Synthesis Quest 1: The Peaked Mountain



LORE: A wrestler with Renal-Ray-Failure is found with a heart rate of 30. The ECG shows massive Peaked-Mountains.

QUEST: Identify the Spark-Plug-Exile (Hyper-K-the-Killer) and execute the first counter-move: Calcium-the-Shield!

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Synthesis Quest 2: The Sour Blood



LORE: A diabetic wrestler arrives unconscious. Bella says pH is 7.0, and the Anion-Gap is exactly 25.

QUEST: Identify the Mystery-Acid (Ketoacids) and diagnose the Missing-Gap syndrome (DKA).

Synthesis Quest 3: The Yellow Pipe-Block



LORE: Wrestler has yellow skin, dark urine, pale stools. Lab reveals High-Direct-Bilirubin and High-Lipase.

QUEST: Identify the Bile-BUILDER-Block (Obstructive Jaundice) triggering the pancreatic Autodigestion-Alarm (Pancreatitis).